Onur Poyraz, PhD

Senior Data Scientist — Probabilistic Machine Learning

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EXPERIENCE

Senior Data Scientist
Vaalia Health
March 2025 – Present Espoo, Finland

- Designed and productionized Hydra-configured forecasting pipelines on Databricks with PySpark and PyTorch Lightning, spanning cohort construction, patient-level stratified splits, cross-validation, feature engineering, and model training
- Developed Transformer-based multi-target time-series models alongside regularized Spark ML baselines, with custom robust losses, output transformations, and calibration evaluation
- Designed causal-inference and counterfactual-simulation workflows for intervention-effect estimation, combining autoregressive baselines, subgroup analysis, and documented DAG assumptions
- Built reusable serverless Spark utilities and production-grade ML governance with MLflow lineage and provenance, automated reporting, pytest, and GitHub Actions

Lead Researcher, Aalto–Nokia Collaboration
Aalto University & Nokia
July 2024 – December 2024 Espoo, Finland

- Led Aalto's research workstream on hardware-aware combinatorial optimization for emerging 6G systems
- Developed an analog–digital Ising machine model for multi-user MIMO detection, translating memristor constraints into algorithm design

Ph.D. Researcher
Aalto University — Machine Learning for Health Group
August 2019 – December 2024 Espoo, Finland

- Developed coupled and mixture hidden Markov models to infer latent interactions and heterogeneous treatment-response patterns in longitudinal healthcare data
- Modeled glucose–insulin dynamics with multi-output hierarchical Gaussian processes, producing uncertainty-aware personalized forecasts
- Developed sparse Kalman filtering methods that scale nonlinear state estimation to high-dimensional physical systems

Data Science Consultant
Boğaziçi University & BKM — Interbank Card Center
June 2018 – June 2019 Istanbul, Turkey

- Delivered a paid university–industry consulting engagement, building Spark-based machine-learning pipelines for large-scale credit-card transaction data
- Developed interpretable fraud-detection and merchant-risk models adopted in BKM's production analytics

PROFILE

Senior data scientist and PhD building production forecasting and causal-ML systems for longitudinal healthcare data. Combines probabilistic modeling with end-to-end ownership across data pipelines, model governance, and deployment.

EDUCATION

Ph.D. in Computer Science
Aalto University
2019 – 2024

Thesis: Flexible Bayesian latent variable modeling of interacting processes in time-series healthcare

M.Sc. in Computational Science and Engineering
Boğaziçi University
2017 – 2019

Thesis: Anomaly Detection in Time Series

B.Sc. in Electrical and Electronics Engineering
Boğaziçi University
2010 – 2016

CORE EXPERTISE

- Time-series forecasting & Transformers
- Causal inference & counterfactuals
- Probabilistic modeling & calibration
- Production ML & model governance
- Longitudinal healthcare data

RECOGNITION

Top 0.006% nationally
Ranked 112th among approximately 2 million participants in Turkey's university entrance examination.

BUALERT engineering team
Designed car electronics, ECU, and energy-management systems for an alternative-energy racing team.

EARLIER EXPERIENCE

Data Scientist

Digital Helix — DeepC

June 2018 – December 2018 | Istanbul, Turkey

- Developed supervised and unsupervised deep-learning workflows for brain-lesion detection in MRI scans
- Worked across image segmentation, classification, and feature regression on high-dimensional medical images

M.Sc. Researcher

Boğaziçi University & Borusan — R&D

February 2017 – April 2018 | Istanbul, Turkey

- Developed unsupervised anomaly-detection models for industrial wind-turbine sensor streams
- Built Bayesian and deep time-series models for real-time detection in noisy multivariate signals

SELECTED RESEARCH IMPACT

Personalized health forecasting
Developed uncertainty-aware models of glucose and insulin dynamics for individualized assessment after bariatric surgery; published in *Communications Medicine* (2025).

Treatment-response modeling
Modeled interactions across MRSA colonization sites to estimate site-specific clearance and treatment effects; published in *Journal of the Royal Society Interface* (2022).

Heterogeneous time-series populations
Introduced mixture coupled hidden Markov models to discover latent response subgroups in multivariate healthcare data; presented at ML4H (2023).

COMMUNITY

- **Scientific Symposium, by Students for Students** — Volunteer, May 2021, Espoo
- **Machine Learning at the Bosphorus İsmail Arı Summer School** — Volunteer, July 2018, İstanbul

TECHNOLOGIES

Core ML
Python, PyTorch, PyTorch Lightning, Spark ML

Data platform
Databricks, PySpark, SQL

MLOps
MLflow, Hydra/OmegaConf, pytest, GitHub Actions

Engineering
uv, Pyrefly

Additional
Stan, R, C/C++, MATLAB, TensorFlow, Julia

LANGUAGES

Turkish
Native

English
Professional fluency

Finnish
Intermediate

PERSONAL INTERESTS